Week 01 Reading: Motivations

Assigned Monday 8/24 and due Wednesday 8/26.

This week only, there is nothing to submit in advance: bring your answers to class and we will use them. From week 2 on, the reading is assigned the week before and your questions are due Sunday night.

Assigned

1. Chapter 1 of Mudelsee (2020), which covers the characteristics of the climate system that, at a broad level, drive our choice of analytical methods. This reading is available through the Rice library.
2. O'Rourke & Burdick (2025), pages 1-6, which describe why the insurance sector estimates hazard from simulated event sets instead of from the historical record alone. This reading is available at nb.com.

Questions to bring Wednesday

Write a few sentences on each. These are not graded and are not submitted; I will ask about them in class and may cold-call on you.

1. Why are you taking this course, and what would make it worth your time?
2. What statistics have you done before? Name the courses or the projects, and say what you actually used rather than what was covered.
3. What is your Earth science or engineering background? What system do you know something about: a river, a coastline, a power grid, a watershed, a city?
4. Is there a dataset or a decision from your own research or work that this course could be useful for? A rough answer is fine.
5. From O'Rourke & Burdick (2025): a catastrophe model produces a distribution of losses, not a single number. Who uses that distribution, and what decision does it inform?
6. Mudelsee (2020) describes the frustration of being unable to reproduce results he read in the literature. What would you need from a paper, or from a colleague, to reproduce their analysis? Why is reproducibility particularly important in climate risk analysis (relative to other fields of engineering)?
7. What is one question about risk, uncertainty, or extremes that you would like this course to answer?

Bibliography

Mudelsee, M. (2020). *Statistical Analysis of Climate Extremes*. Cambridge, United Kingdom ;: Cambridge University Press.

O'Rourke, C., & Burdick, J. (2025). *Catastrophe Modeling 101* (White Paper). Retrieved from https://www.nb.com/